

```
#Experiment 2: Support Vector Machine & Random Forest
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
```

```
iris=load_iris()

X=iris.data
y=iris.target
```

```
from sklearn.model_selection import train_test_split

X_train,X_test,y_train,y_test=train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
from sklearn.svm import SVC

svm=SVC(kernel='linear')

svm.fit(X_train,y_train)

svm_pred=svm.predict(X_test)
```

```
from sklearn.metrics import accuracy_score

print("SVM Accuracy:",accuracy_score(y_test,svm_pred))
```

```
SVM Accuracy: 1.0
```

```
from sklearn.ensemble import RandomForestClassifier

rf=RandomForestClassifier(random_state=42)

rf.fit(X_train,y_train)

rf_pred=rf.predict(X_test)
```

```
print("Random Forest Accuracy:",accuracy_score(y_test,rf_pred))
```

```
Random Forest Accuracy: 1.0
```

```
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_test,rf_pred))
```

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

```
from sklearn.metrics import classification_report

print(classification_report(y_test,svm_pred))
```

```

              precision    recall  f1-score   support

     0         1.00      1.00      1.00         10
     1         1.00      1.00      1.00          9
     2         1.00      1.00      1.00         11

   accuracy          1.00
  macro avg          1.00
 weighted avg          1.00
```

```
print(classification_report(y_test,rf_pred))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```
accuracies=[
    accuracy_score(y_test,svm_pred),
    accuracy_score(y_test,rf_pred)
]

models=['SVM','Random Forest']

plt.bar(models,accuracies)

plt.title("Accuracy Comparison")

plt.ylabel("Accuracy")

plt.show()
```

